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COURSE NAME: ANTENNA AND WAVE PROPAGATION

EX NO: 9A

SIMULATION OF GROUND WAVE AND SKY WAVE PROPAGATION USING MATLAB

DATE: 11/07/26

OBJECTIVES

- i) To simulate Ground Wave Propagation and analyze the variation of received signal strength with distance using MATLAB.
 - ii) To simulate Sky Wave Propagation and determine the skip distance and Maximum Usable Frequency (MUF) using MATLAB.
 - iii) To compare Ground Wave and Sky Wave Propagation characteristics through graphical analysis and evaluate their communication coverage.
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OBJECTIVE 1

SIMULATION OF GROUND WAVE PROPAGATION

PROCEDURE

1. Open MATLAB and create a new script.
 2. Define the transmission frequency and distance range.
 3. Generate the ground wave propagation model.
 4. Calculate attenuation with distance.
 5. Plot received signal strength versus distance.
 6. Analyze the propagation characteristics.
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FORMULA

Ground Wave Electric Field Strength

$$E = 300\sqrt{P_t} / d$$

Where:

- E = Electric field strength (mV/m)
- P_t = Transmitted power (kW)
- d = Distance (km)

Path Loss

$$L = 20 \log(d) + 20 \log(f) + 32.44$$

RESULT

The variation of ground wave signal strength with distance was successfully simulated.

OBJECTIVE 2

SIMULATION OF SKY WAVE PROPAGATION

PROCEDURE

1. Define the ionospheric layer height.
 2. Define the operating frequency.
 3. Calculate the skip distance.
 4. Determine the Maximum Usable Frequency (MUF).
 5. Simulate signal reflection from the ionosphere.
 6. Plot the propagation path.
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FORMULA

Skip Distance

$$D_s = 2h \tan \theta$$

Maximum Usable Frequency

$$MUF = f_c \sec \theta$$

Where:

- h = Ionospheric height
 - θ = Angle of incidence
 - f_c = Critical frequency
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RESULT

The skip distance and Maximum Usable Frequency (MUF) for sky wave propagation were successfully determined.

OBJECTIVE 3

COMPARATIVE ANALYSIS OF GROUND WAVE AND SKY WAVE PROPAGATION

PROCEDURE

1. Simulate ground wave propagation.
 2. Simulate sky wave propagation.
 3. Plot both propagation characteristics.
 4. Compare communication ranges.
 5. Analyze propagation losses.
 6. Interpret the results.
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FORMULA

Propagation Delay

$$t = d / c$$

Where:

$$c = 3 \times 10^8 \text{ m/s}$$

RESULT

Ground wave and sky wave propagation characteristics were successfully compared and analyzed.

OVERALL MATLAB PROGRAM

```
clc;
clear;
close all;

%% Objective 1: Ground Wave Propagation
Pt = 10;           % Transmitted Power (kW)
d = 1:1:500;       % Distance (km)
E_ground = (300*sqrt(Pt))./d;

%% Objective 2: Sky Wave Propagation
h = 250;           % Ionosphere Height (km)
theta = 60;        % Angle of Incidence (degrees)
```

```

fc = 5;                % Critical Frequency (MHz)

Skip_Distance = 2*h*tand(theta);

MUF = fc*sec(deg2rad(theta));

%% Objective 3: Comparative Analysis

Sky_Range = Skip_Distance*ones(size(d));

%% Display Results

fprintf('\nGROUND WAVE AND SKY WAVE ANALYSIS\n');

fprintf('-----\n');

fprintf('Transmitted Power = %.2f kW\n', Pt);

fprintf('Ionospheric Height = %.2f km\n', h);

fprintf('Skip Distance = %.2f km\n', Skip_Distance);

fprintf('Maximum Usable Frequency (MUF) = %.2f MHz\n', MUF);

%% Plot Results

figure('Name', ...

      'Ground Wave and Sky Wave Propagation', ...

      'NumberTitle','off');

% Ground Wave Propagation

subplot(2,2,1);

plot(d,E_ground,'LineWidth',2);

grid on;

xlabel('Distance (km)');

ylabel('Field Strength (mV/m)');

title('Ground Wave Propagation');

% Maximum Usable Frequency

subplot(2,2,2);

bar(MUF);

grid on;

title('Maximum Usable Frequency');

ylabel('Frequency (MHz)');

set(gca,'XTickLabel',{'MUF'});

```

```

% Skip Distance

subplot(2,2,3);

bar(Skip_Distance);

grid on;

title('Skip Distance');

ylabel('Distance (km)');

set(gca,'XTickLabel',{'Sky Wave'});

% Ground Wave Field Strength

subplot(2,2,4);

plot(d,E_ground,'LineWidth',2);

grid on;

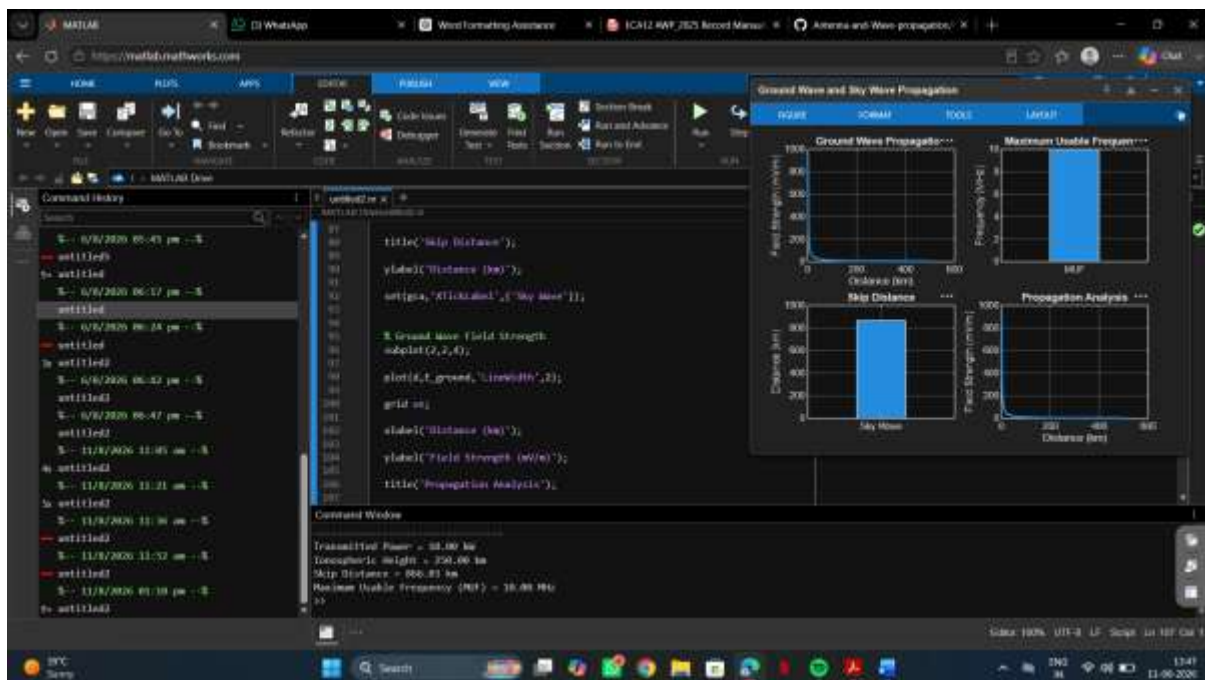
xlabel('Distance (km)');

ylabel('Field Strength (mV/m)');

title('Propagation Analysis');

```

OUTPUT :



FINAL RESULT

Ground wave and sky wave propagation were successfully simulated using MATLAB. The variation of ground wave field strength with distance was analyzed, while the skip distance and Maximum Usable Frequency (MUF) of sky wave propagation were calculated. The propagation characteristics were graphically analyzed and compared successfully.